

Finite-quotient and eventual-block complementary spectra

A partial result on Question (Q4) of Gabardo–Lai

Candidate proof packet for arXiv:1311.2209

13 August 2026

Status. Candidate partial result, likely valid pending human review. The unrestricted type-II question is not claimed solved.

1 Source question

Gabardo and Lai [1] consider integers $N_n \geq 2$, put $P_0 = 1$ and $P_n = N_1 \cdots N_n$, and define

$$\eta_n = \frac{1}{N_n} \sum_{r=0}^{N_n-1} \delta_{r/P_n}.$$

The associated type-II natural pair is

$$\mu = \bigast_{j \geq 1} \eta_{2j-1}, \quad \nu = \bigast_{j \geq 1} \eta_{2j}, \quad \mu * \nu = \mathcal{L}_{[0,1]}.$$

On page 22, Question (Q4) asks whether, for every spectrum $\Lambda_m u$ of μ , there is a spectrum Λ_ν of ν such that $\Lambda_\mu \oplus \Lambda_\nu = \mathbb{Z}$. The paper notes that the remaining case is type II.

posing an absolutely continuous (w.r.t. Lebesgue) measures with Fourier frames? Conversely, is it true that all measures admitting Fourier frames are constructed in this way?

Given a spectral measure μ , another important issue is to classify its spectrum. This question has been studied for Lebesgue measures and some Cantor measures in [LRW, DHS, DHL]. However, there is no satisfactory answer when the measure is singular. The tiling statement of Theorem 1.1, suggests a possible answer.

(Q4): Let μ and ν be a natural complementary pair of $\mathcal{L}_{[0,1]}$. Let also Λ_μ be a spectrum for $L^2(\mu)$, does there exist a spectrum Λ_ν for $L^2(\nu)$ such that $\Lambda_\mu \oplus \Lambda_\nu = \mathbb{Z}$?

It is not difficult to prove that **(Q4)** actually holds for type I decompositions. The remaining challenge is to answer the question for type II decompositions.

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2 Result and proof intuition

At finite depth, the Fourier zeros have alternating mixed-radix layers. Every difference in a spectrum of the odd head lies in an odd layer; every difference in a spectrum of the even head lies in an

even layer. Hence two head sums cannot collide modulo P_{2k} . Their cardinalities multiply to P_{2k} , so the sums give every residue exactly once. This works for arbitrary finite spectra, not just the digit spectra displayed in the source paper.

For an infinite spectrum with a genuine head–tail product, the finite head sum may represent residue r by $r + P_{2k}c_r$ rather than by r itself. An exact tail tiling absorbs the integer carry c_r . This proves a recursive reduction and, in particular, Question (Q4) for arbitrary finite spectral perturbations followed by a canonical tail.

The argument does not pass automatically to a general spectral tree. The finite decompositions always converge in the mixed-radix profinite completion, but the resulting paths need not terminate as ordinary integers; moreover, a maximal orthogonal tree need not be a complete spectrum [2]. This is the remaining obstruction.

3 Finite complementary spectra

Define the finite heads

$$\mu_k = \eta_1 * \eta_3 * \cdots * \eta_{2k-1}, \quad \nu_k = \eta_2 * \eta_4 * \cdots * \eta_{2k}.$$

For a finite measure, a “spectrum” below is a frequency set whose exponentials form an orthonormal basis of its finite-dimensional L^2 space.

Lemma 1 (Fourier-zero layers). *The real zero set of $\hat{\eta}_n$ is*

$$Z(\hat{\eta}_n) = P_{n-1}\mathbb{Z} \setminus P_n\mathbb{Z}.$$

Consequently,

$$\begin{aligned} Z(\hat{\mu}_k) &= \bigcup_{j=1}^k (P_{2j-2}\mathbb{Z} \setminus P_{2j-1}\mathbb{Z}), \\ Z(\hat{\nu}_k) &= \bigcup_{j=1}^k (P_{2j-1}\mathbb{Z} \setminus P_{2j}\mathbb{Z}). \end{aligned}$$

Proof. The Fourier transform of η_n is the geometric sum

$$\hat{\eta}_n(\xi) = \frac{1}{N_n} \sum_{r=0}^{N_n-1} e^{-2\pi i r \xi / P_n}.$$

It vanishes exactly when ξ/P_n is a nonintegral N_n -th root frequency, equivalently when $\xi \in P_{n-1}\mathbb{Z} \setminus P_n\mathbb{Z}$. The remaining claims follow by taking products. $\square$

Theorem 2 (Arbitrary finite spectra tile the quotient). *Let L_0 be any spectrum of μ_k and C_0 any spectrum of ν_k . Choose $\ell_0 \in L_0, c_0 \in C_0$ and normalize $L = L_0 - \ell_0, C = C_0 - c_0$. Then*

$$L \oplus C = H_k,$$

where H_k is a complete system of residues modulo P_{2k} . Equivalently, the addition map

$$L \times C \longrightarrow \mathbb{Z}/P_{2k}\mathbb{Z}, qquad (\ell, c) \longmapsto \ell + c$$

is bijective.

Proof. The normalization makes both spectra contain zero. Orthogonality and Lemma 1 then imply $L, C \subset \mathbb{Z}$. If $\ell \neq \ell'$ in L , the difference $\ell - \ell'$ belongs to a unique odd layer below $2k$. Likewise, if $c \neq c'$ in C , then $c - c'$ belongs to an even layer below $2k$.

Suppose

$$\ell + c \equiv \ell' + c' \pmod{P_{2k}}.$$

If both differences are nonzero, then $\ell - \ell' = (c' - c) + mP_{2k}$ for some $m \in \mathbb{Z}$. Adding a multiple of P_{2k} cannot change the first failed divisibility layer below $2k$. The left side therefore has an odd layer and the right side an even layer, a contradiction. If one difference is zero, the other is congruent to zero modulo P_{2k} . This is also impossible for distinct frequencies: the displayed zero layers contain no multiple of P_{2k-1} for μ_k and no multiple of P_{2k} for ν_k . Thus the addition map is injective.

Finally,

$$|L| = N_1 N_3 \cdots N_{2k-1}, \quad |C| = N_2 N_4 \cdots N_{2k},$$

because these are the support cardinalities of the finite measures. Hence $|L||C| = P_{2k}$, and injectivity is bijectivity. $\square$

In particular, one may take

$$C = -\bigoplus_{j=1}^k P_{2j-1} \{0, 1, \dots, N_{2j} - 1\},$$

the negative canonical even-head spectrum from [1].

4 A head–tail spectrum lemma

For $P = P_{2k}$, rescale the tails by P :

$$\bar{\mu}^{(k)} = (x \mapsto Px)_{\#} \ast_{j>k} \eta_{2j-1}, \quad \bar{\nu}^{(k)} = (x \mapsto Px)_{\#} \ast_{j>k} \eta_{2j}.$$

These are again a natural type-II pair, now for the tail sequence $N_{2k+1}, N_{2k+2}, \dots$

Lemma 3 (Product spectra with a finite head). *If L is a spectrum of μ_k and U a spectrum of $\bar{\mu}^{(k)}$, then*

$$L \oplus PU$$

is a spectrum of μ . Similarly, if C is a spectrum of ν_k and V a spectrum of $\bar{\nu}^{(k)}$, then $C \oplus PV$ is a spectrum of ν .

Proof. We prove the odd case. The digit expansion makes addition from the head and tail supports one-to-one outside the usual null set of ambiguous endpoint expansions. It therefore identifies $L^2(\mu)$ with $L^2(\mu_k) \otimes L^2(\bar{\mu}^{(k)})$.

Write a point as $a + b/P$, with a in the head support and b in the rescaled tail support. Since $a \in P_{2k-1}^{-1}\mathbb{Z}$ and $P = P_{2k-1}N_{2k}$, one has $Pa \in \mathbb{Z}$. Thus

$$e_{\ell+Pu}(a + b/P) = e_{\ell}(a) e_{u+\ell/P}(b).$$

For each fixed $\ell \in L$, the set $\{e_{u+\ell/P} : u \in U\}$ is merely a common frequency translate of the orthonormal basis $E(U)$ in the tail space. The displayed functions therefore form an orthonormal basis of the tensor product: the first factors run through the basis $E(L)$, and inside each first-factor summand the second factors form a full tail basis. The even case is identical. $\square$

5 The eventual block-product theorem

Theorem 4 (Recursive tail reduction). *Fix $k \geq 1$ and $P = P_{2k}$. Let L, C be arbitrary normalized spectra of μ_k, ν_k , meaning that $0 \in L \cap C$. Suppose U, V are spectra of the renormalized tails $\bar{\mu}^{(k)}, \bar{\nu}^{(k)}$ and*

$$U \oplus V = \mathbb{Z}.$$

Then

$$\Lambda = L \oplus PU, \text{ and } \Gamma = C \oplus PV$$

are spectra of μ, ν , respectively, and

$$\Lambda \oplus \Gamma = \mathbb{Z}.$$

The conclusion is invariant under opposite common translations of Λ and Γ .

Proof. Spectrality follows from Lemma 3. By Theorem 2, $H = L \oplus C$ contains exactly one representative h_r of each residue $r \in \{0, \dots, P-1\}$. Write $h_r = r + Pq_r$ with $q_r \in \mathbb{Z}$.

Given $z = r + Pq \in \mathbb{Z}$, choose the unique head pair $h_r = \ell + c$. Since $U \oplus V = \mathbb{Z}$, choose the unique (u, v) satisfying $u + v = q - q_r$. Then

$$z = (\ell + Pu) + (c + Pv).$$

Conversely, any two such representations have head sums congruent modulo P , hence the same r and, by the directness in Theorem 2, the same ℓ, c . Tail directness then gives the same u, v . This proves the exact direct sum.

If the prescribed spectrum is translated by t , translate the constructed complement by $-t$; frequency translation preserves spectrality and leaves the sum equal to $\mathbb{Z}$. $\square$

Corollary 5 (Eventually canonical spectra). *Question (Q4) has an affirmative answer for every prescribed spectrum of the form*

$$t + L \oplus P_{2k}U_{\text{can}},$$

where L is an arbitrary normalized finite spectrum of μ_k and U_{can} is the canonical odd spectrum of the renormalized tail. More generally, it holds for every eventual block-product spectrum whose terminal tail pair already has complementary spectra.

Proof. Gabardo–Lai prove that the canonical odd tail spectrum and the negative canonical even tail spectrum have direct sum $\mathbb{Z}$ [1, Section 3]. Apply Theorem 4, and iterate it for the more general statement. $\square$

6 Verification and limitations

The accompanying verifier exhaustively enumerates all normalized odd-head spectra in the binary depth-two quotient and in the mixed-radix cases $(2, 3, 2, 2)$, $(2, 2, 3, 2)$, and $(3, 2, 2, 2)$. In each case it checks direct tiling by the negative canonical even spectrum. This is a guard against layer and modulus mistakes, not part of the proof.

The unrestricted part of (Q4) remains open in this packet. An arbitrary spectrum may have a tree structure with unbounded delays before a branch terminates. Theorem 2 supplies compatible decompositions at every finite quotient, but compactness only produces a profinite branch. It does not prove that the branch represents an ordinary integer, nor that a resulting maximal orthogonal even tree is complete. Eight focused upgrade routes, including a greedy complement construction and a proposed analytic completeness transfer, did not remove this obstruction.

A bounded search through 13 August 2026 used the exact question wording, source citation graph, and later Cantor/Cantor–Moran spectrum literature. No explicit all-spectra resolution of (Q4) was found. The novelty assessment is therefore moderate and requires expert confirmation.

7 Human-review recommendation

The result is suitable for expert review as a substantial partial theorem. The most important checks are the zero-layer convention in Lemma 1, the injectivity modulo P_{2^k} in Theorem 2, and the tensor-basis identification in Lemma 3. Reviewers should preserve the explicit limitation: no claim is made for a general nonterminating spectral tree.

References

- [1] J.-P. Gabardo and C.-K. Lai, *Spectral Measures Associated with the Factorization of the Lebesgue Measure on a Set via Convolution*, J. Fourier Anal. Appl. 20 (2014), 453–475; arXiv:1311.2209.
- [2] X.-R. Dai, *Spectra of Cantor measures*, Math. Ann. 366 (2016), 1621–1647; arXiv:1401.4630.